

Information Technology

Spring Semester
2025 - 2026

» Lec. 1: Introduction to Pattern
Recognition

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EELU

الجامعة المصرية للتعليم الإلكتروني
THE EGYPTIAN E-LEARNING UNIVERSITY

Student assessments + Assessment Grades

10 Project

10 Quizzes (2 quizzes each 5)

30 Midterm Exam

50 Final Exam

Section Attendance (minus)

Pattern Recognition

TEXT BOOKS:

1. Pattern Classification (2nd ed.), David G. Stork

REFERENCE BOOKS:

1. An Introduction to Pattern Recognition and Machine Learning - Paul Fieguth

Introduction to Pattern Recognition

1. What is Pattern Recognition?

Machine-driven Data Analysis

- At the age of 5, most children can recognize digits and letters – small characters, large characters, handwritten, machine printed, or rotated – all easily recognized by the young.
- In most instances, the best pattern recognizers are **humans**, yet we do not understand how humans recognize patterns.

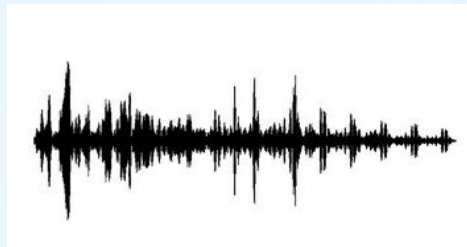
1. What is Pattern Recognition?

Machine-driven Data Analysis

- The rapidly growing amount of training data generated is making **it impossible for humans to interpret** and driving the **need for machines** to be able to identify patterns *quickly and accurately*.
- The ability to automate the recognition of patterns and regularities in data has many applications, from facial recognition software to tumor detection.

1. What is Pattern Recognition?

- **What is pattern?** A pattern can be any entity of interest that one needs to recognize and identify.
- **Pattern recognition** is the ability of machines to identify patterns in data and then use those patterns to make decisions or predictions using **computer algorithms**.
- It's a vital component of modern artificial intelligence (AI) systems.



1. What is Pattern Recognition?

- Pattern recognition is **defined as**:
 - The study of how machines can **observe** the environment, **learn** to distinguish various patterns of interest from their background, and **make logical decisions** about the categories of the patterns.
 - During recognition, the given objects are assigned to a specific category.

1. What is Pattern Recognition?

- There are several **definitions** of Pattern Recognition, as it is a constantly evolving and broad field.
- An early definition of pattern recognition defines it as “a classification of input data via extraction of important features from a lot of noisy data” (1978, Thomas Gonzalez).

1. What is Pattern Recognition?

- **Another definition** describes pattern recognition as “**a scientific discipline whose aim is the classification of the objects into a lot of categories or classes. Pattern recognition is also an integral part of most machine intelligence systems built for decision making**” (2003, Sergio Theodoridis).

1. What is Pattern Recognition?

- In general, pattern recognition can be described as **an information reduction**, information mapping, or information labeling process. In computer science, pattern recognition refers to the process of matching information already stored in a database with incoming data based on their attributes.

1. What is Pattern Recognition?

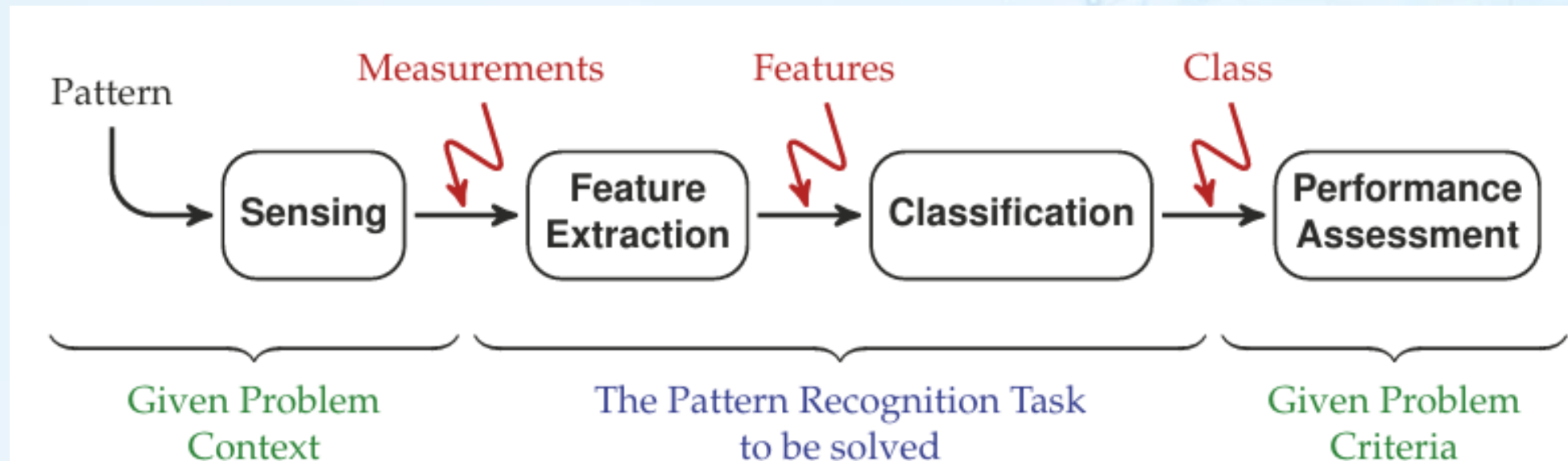
Pattern recognition task	Possible classes
What is in front of you as you walk?	Door vs. Window Sidewalk vs. Road
What music are you listening to?	Familiar or Unfamiliar Genre (Rock, Classical, ...) Name of Composer or Group
Is the traffic intersection safe to cross?	Green vs. Red light Pedestrian Walk vs. Stop Car Present vs. Not Present
Reading a page in a textbook	Letters of the Alphabet Text vs. Graphics Languages
You smell something in your apartment ...	Cookies finished baking? (Yes/No) Is something burning? (Yes/No) Wet dog/Skunk/Dead mouse/...

1. What is Pattern Recognition?

- As a human experience, **pattern recognition** refers to a **perceptual process** in which some **form of sensory input** is sensed, analyzed, and recognized (classified), either subconsciously (by instinct) or consciously (based on previous experience).
- Patterns may be presented in any **sensory modality**: vision, hearing, touch, taste, or smell.

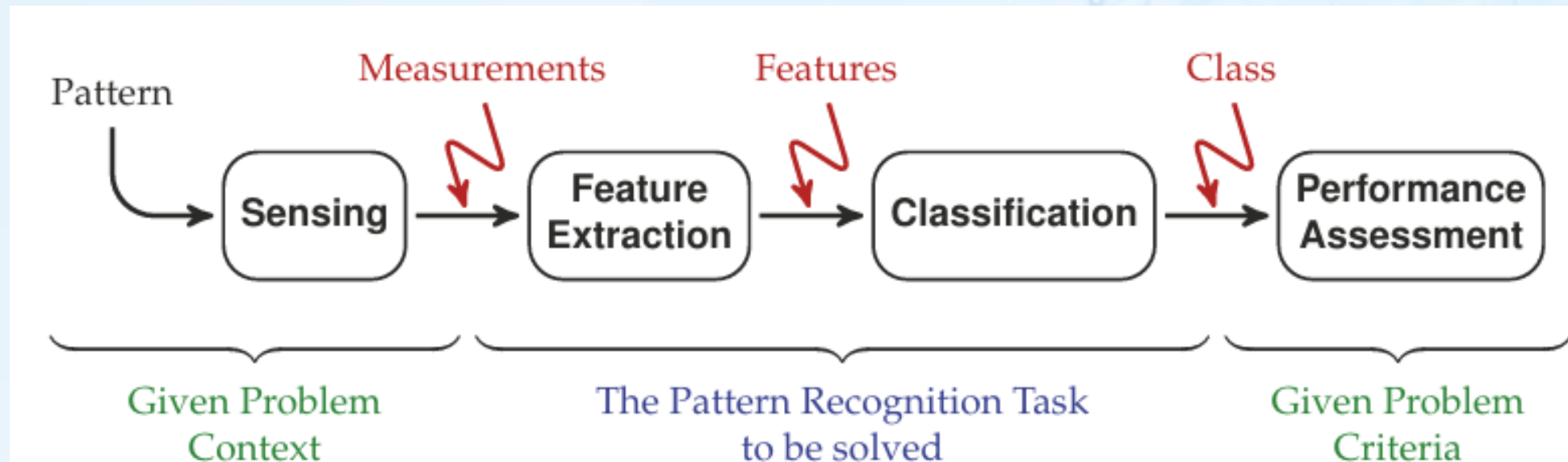
1. What is Pattern Recognition?

- As a **technical discipline**, **pattern recognition** refers to a process in which an **input object** is measured, analyzed, and classified by a machine as being more or less similar to some class in a set of classes.



1. What is Pattern Recognition?

- **PATTERN RECOGNITION:** A pattern is sensed, giving rise to measurements, from which feature are extracted and given to a classifier, whose job it is to select the class associated with the sensed pattern.



1. What is Pattern Recognition?

Example 1: Pattern Recognition of Text.

When you read a printed character on this page, your eye images the character on your retina, where it is sensed and converted into a neural representation which is subsequently analyzed in your brain. If your memory indicates prior experience with the character, or with a symbol sufficiently similar to it, you perceive or recognize the symbol, associating a label and meaning with it. What appears to us to be a relatively simple process becomes ever more impressive as we attempt to design a machine which might accomplish the same task, known as optical character recognition (OCR). What features might a machine use in recognizing text?

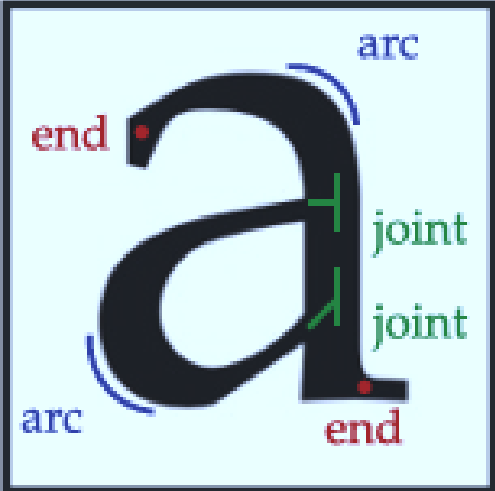
1. What is Pattern Recognition?

Example 1- A: Pattern Recognition of Text.

Pattern	Attributes to Measure	Measurements	Strengths and Weaknesses
"a" → 		Vector of pixel values	Fast, easy, explicit Sensitive to changes in font style, size, and rotation

1. What is Pattern Recognition?

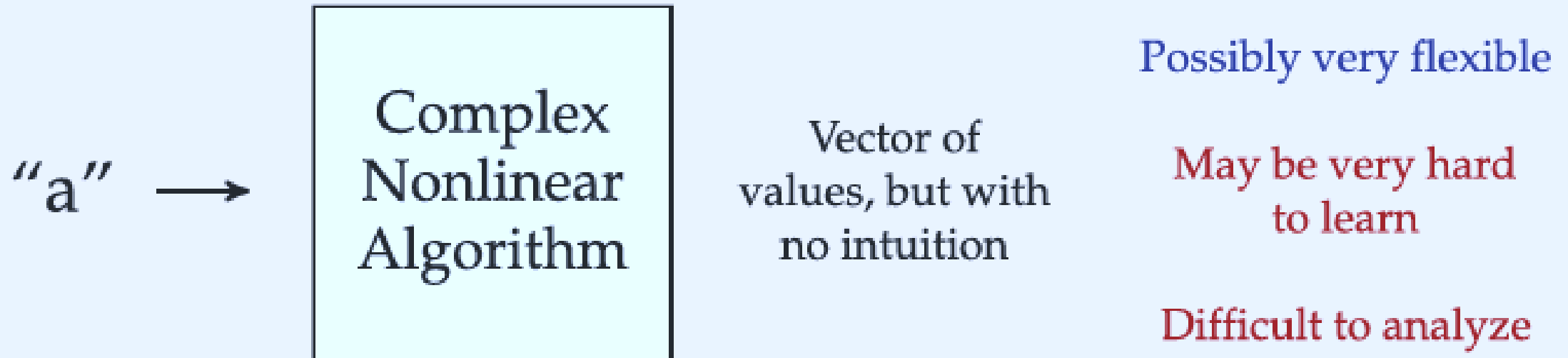
Example 1- B: Pattern Recognition of Text.

Pattern	Attributes to Measure	Measurements	Strengths and Weaknesses
"a" →		Vector of shape properties	Robust to changes in size and rotation Complicated features to extract

1. What is Pattern Recognition?

Example 1- C: Pattern Recognition of Text.

Pattern	Attributes to Measure	Measurements	Strengths and Weaknesses
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- It can be challenging to train or adjust the algorithm because of its complexity.
- Understanding why the algorithm makes specific decisions can be tough, due to its non-linear and intricate nature.

1. What is Pattern Recognition?

- For a machine needing to deal with **printed characters** in only a single font, a simple classification scheme might sense an array of pixel intensity values (**Example 1- A**) and compare it to arrays stored in memory corresponding to known characters.

1. What is Pattern Recognition?

- Such a template matching approach is simple and fast, but will be subject to a variety of limitations: any **distortion, translation, rotation, scale** change, or even **lighting variations** will affect the degree to which the given measurement matches a template.

1. What is Pattern Recognition?

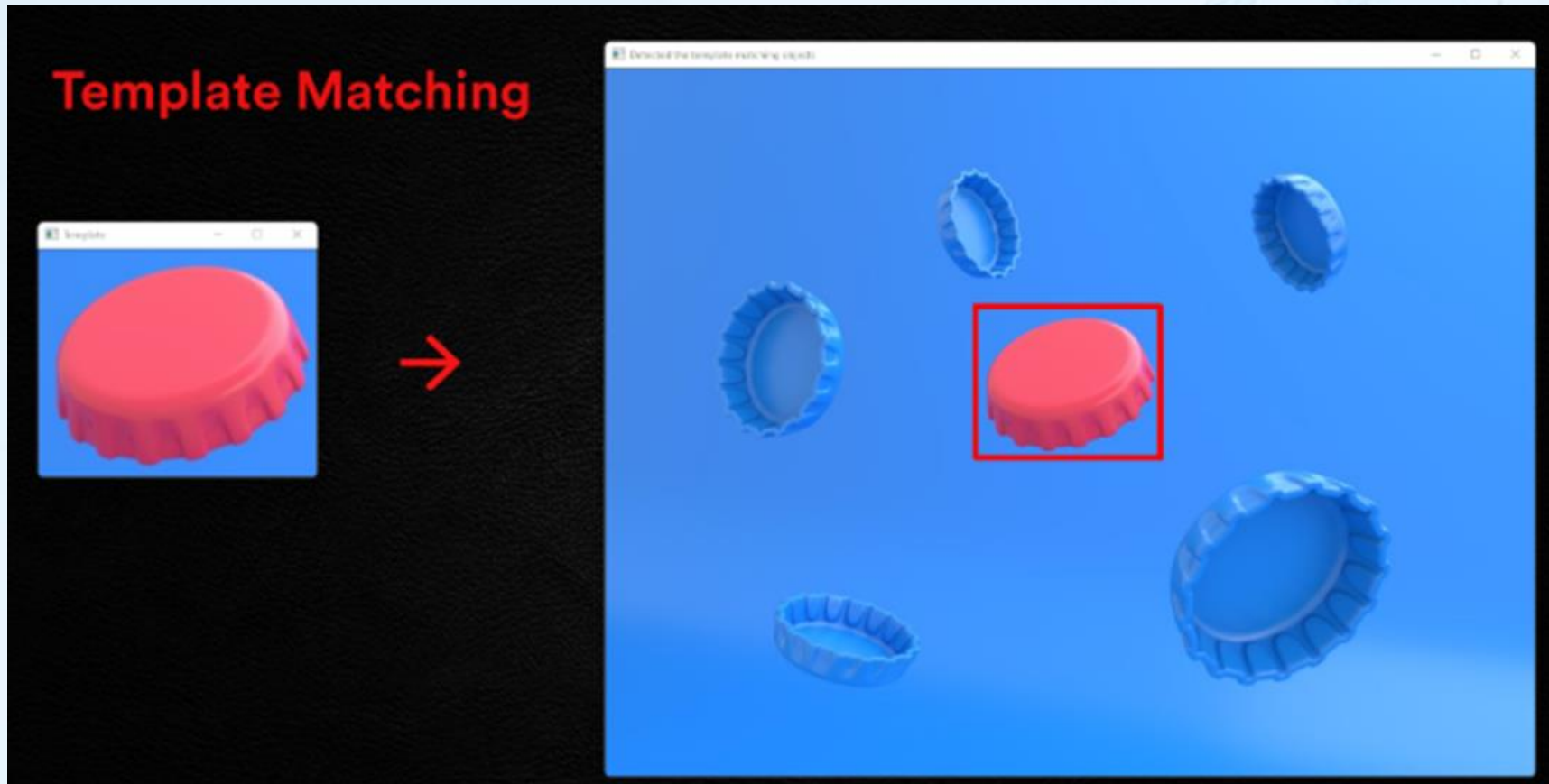
- **Template matching** is one of the simplest and earliest approaches to pattern recognition.
- **Matching** is a generic operation that is used to determine the similarity between two entities of the same type.
- Therefore, template-matching models try to **discover similarities** in a sample based on a reference template.

1. What is Pattern Recognition?

- Hence, the template matching technique is commonly used in digital image processing for detecting small sections of an image that match a template image.
- Typical real-world **examples** are:
 - Medical Image Processing,
 - Quality Control in Manufacturing,
 - Robot Navigation, or Face Recognition.

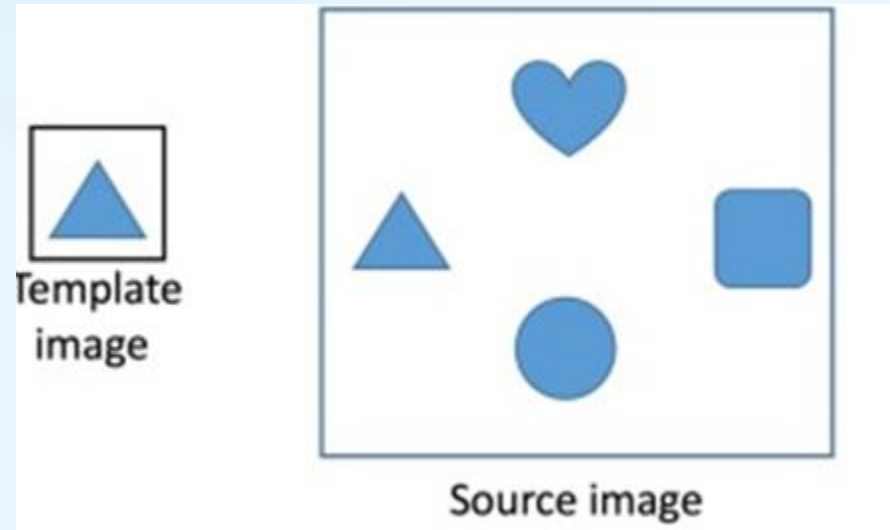
1. What is Pattern Recognition?

Template Matching



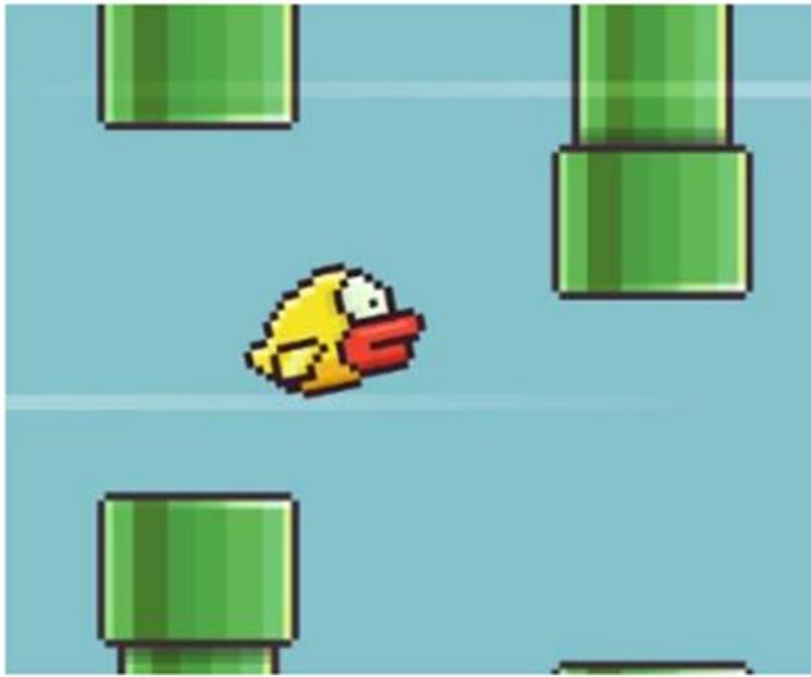
1. What is Pattern Recognition?

Template Matching



1. What is Pattern Recognition?

Template Matching



Large Image



Template Image

1. What is Pattern Recognition?

- If the machine must accept a variety of fonts, as humans would in reading, a memorized template matching scheme becomes impractical, since an unreasonable number of templates would need to be stored and compared.

1. What is Pattern Recognition?

- **An alternate approach** is to seek distinctive character features (**Example 1- B**), such as strokes, arcs, loops etc.
- It might also be possible to train a black-box strategy (**Example 1- C**), whereby some complex nonlinear function is learned, producing features which no longer have any intuitive (Common-sense) meaning regarding recognizing characters, but which turn out to be effective features in classifying typed characters.



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THANK YOU FOR WATCHING

QUESTIONS?

